First version

Sets and parameters:

D ordered set of all days of the scheduling period, where $d = 1$ is the first day and $d = |D|$ is the last day of the scheduling period

T ordered set of all possible timeslots in any day, where $t = 1$ is the first timeslot of the day and $t = |T|$ is the last time slot of the day

$T_d \subset T$ set of timeslots that must be covered in day $d \in D$

W set of all workers $w \in W$

S set of all skills $s \in S$

$S_w \subset S$ set of skills of worker $w \in W$

H_{wd} set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$

α_{dts} minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

Variables:

x_{wdh} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$

y_{wdts} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$

z_{dts} integer variable with the number of workers below the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

ObjectiveFunction1(α_{dts}): **trying to fulfil as much as possible the minimum number α_{dts} of workers in all days, timeslots and skills**

$$\text{Minimize} \sum_{d \in D} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \qquad (1)$$

The aim is to minimize the sum of the number of workers below the required value α_{dts} in all days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$.

Constraints:

$$\sum_{h \in H_{wd}} x_{wdh} \leq 1 \qquad , w \in W, d \in D \qquad (2)$$

With constraints (2), each worker $w \in W$ can be assigned on each day $d \in D$ with at most one daily working assignment $h \in H_{wd}$.

$$\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} \delta_{wdht}\, x_{wdh} \qquad , w \in W, d \in D, t \in T_d \qquad (3)$$

With constraints (3), each worker $w \in W$ assigned on day $d \in D$ with a daily working assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$, must be assigned in timeslot t with one of its skills S_w.

$$\sum_{d=d'}^{d'+5} \sum_{h \in H_{wd}} x_{wdh} \leq 5 \qquad , w \in W, d' = 1, 2, \dots, |D| - 5 \qquad (4)$$

With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive days.

$$z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts} \qquad , d \in D, t \in T_d, s \in S \qquad (5)$$

With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$.

Second version (adding competence levels of workers on each of their skills)

Sets and parameters:

D	ordered set of all days of the scheduling period, where $d = 1$ is the first day and $d = \lvert D \rvert$ is the last day of the scheduling period
T	ordered set of all possible timeslots in any day, where $t = 1$ is the first timeslot of the day and $t = \lvert T \rvert$ is the last time slot of the day
$T_d \subset T$	set of timeslots that must be covered in day $d \in D$
W	set of all workers $w \in W$
S	set of all skills $s \in S$
L	ordered set of all competence levels among all skills, where $l = 1$ represents the highest level and $l = \lvert L \rvert$ represents the lowest level
$S_w \subset S$	set of skills of worker $w \in W$; for each $s \in S_w$, parameter $l_{ws} \in L$ indicates the competence level of worker $w \in W$ in skill $s \in S_w$
H_{wd}	set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$
α_{dts}	minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
β_{dtsl}	minimum number of workers required on day $d \in D$, timeslot $t \in T_d$, skill $s \in S$ and competence level $l \in L$

Variables:

x_{wdh}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$
y_{wdts}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$
y'_{wdtsl}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ and competence level $l \in L$
z_{dts}	integer variable with the number of workers below the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
z'_{dtsl}	integer variable with the number of workers below the required value β_{dtsl} on day $d \in D$, timeslot $t \in T_d$, skill $s \in S$ and competence level $l \in L$

ObjectiveFunction1(α_{dts}): **trying to fulfil as much as possible the minimum number α_{dts} of workers in all days, timeslots and skills**

$$\text{Minimize} \sum_{d \in D} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \qquad (1)$$

The aim is to minimize the sum of the number of workers below the required value α_{dts} in all days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$.

ObjectiveFunction2(s, l, β_{dtsl}): **for a given skill s and competence level l, trying to fulfil as much as possible the minimum number β_{dtsl} of workers in all days and timeslots**

$$\text{Minimize} \sum_{d\in D} \sum_{t\in T_d} z'_{dtsl} \qquad (1')$$

The aim is to minimize the sum of the number of workers below the required value β_{dtsl} in all days $d \in D$ and timeslots $t \in T_d$ for a given skill $s \in S$ and competence level $l \in L$.

ObjectiveFunction3: **trying to obtain the best possible average competence level**

$$\text{Minimize} \sum_{w\in W} \sum_{d\in D} \sum_{t\in T_d} \sum_{s\in S} \sum_{l\in L} (l \times y'_{wdtsl}) \qquad (1'')$$

The aim is to minimize the average competence level assigned to all workers $w \in W$ in all days $d \in D$, all timeslots $t \in T_d$ and all skills $s \in S$ assuming linear competence level weights (i.e., competence level l has weight l).

Constraints:

$$\sum_{h\in H_{wd}} x_{wdh} \leq 1 \qquad , w \in W, d \in D \qquad (2)$$

With constraints (2), each worker $w \in W$ can be assigned on each day $d \in D$ with at most one daily working assignment $h \in H_{wd}$.

$$\sum_{s\in S_w} y_{wdts} = \sum_{h\in H_{wd}} \delta_{wdht}\, x_{wdh} \qquad , w \in W, d \in D, t \in T_d \qquad (3)$$

With constraints (3), each worker $w \in W$ assigned on day $d \in D$ with a daily working assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$, must be assigned in timeslot t with one of its skills S_w.

$$\sum_{d=d'}^{d'+5} \sum_{h\in H_{wd}} x_{wdh} \leq 5 \qquad , w \in W, d' = 1, 2, \ldots, |D| - 5 \qquad (4)$$

With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive days.

$$z_{dts} + \sum_{w\in W} y_{wdts} \geq \alpha_{dts} \qquad , d \in D, t \in T_d, s \in S \qquad (5)$$

With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$.

$$y'_{wdtsl} = \begin{cases} y_{wdts} & , l = l_{ws} \\ 0 & , l \neq l_{ws} \end{cases} \qquad , w \in W, d \in D, t \in T_d, s \in S_w \qquad (6)$$

With constraints (6), each worker $w \in W$ that works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ is assigned with its competence level l_{ws} and the variables associated to all other competence levels are set to zero.

$$z'_{dtsl} + \sum_{w\in W} y'_{wdtsl} \geq \beta_{dtsl} \qquad , d \in D, t \in T_d, s \in S, l \in L \qquad (7)$$

With constraints (7), the value of variable z'_{dtsl} is at least the number of workers below the required value β_{dtsl} for each day $d \in D$, each timeslot $t \in T_d$, each skill $s \in S$ and each competence level $l \in L$. (NOTE: Constraints (7) are required only if ObjectiveFunction2 is used).

Third version (adding variable days-off and closed days)

Sets and parameters:

D	ordered set of all days of the scheduling period, where $d = 1$ is the first day and $d = \|D\|$ is the last day of the scheduling period
$D_o \subset D$	set of shop open days
$\bar{D} \subset D$	set of days $d \in D$ such that all days in the interval from d up to $d + 5$ (inclusive) belong to D_o
T	ordered set of all possible timeslots in any day, where $t = 1$ is the first timeslot of the day and $t = \|T\|$ is the last time slot of the day
$T_d \subset T$	set of timeslots that must be covered in day $d \in D$
W	set of all workers $w \in W$
K	ordered set of weeks of the scheduling period, where $k = 1$ is the first week and $k = \|K\|$ is the last week of the scheduling period
$D_k \subset D_o$	set of open days of week $k \in K$
$D_{wk} \subset D_k$	set of preferable variable days-off of worker $w \in W$ in week $k \in K$
$U_{wk} \subset D_k$	set of unavailable days of worker $w \in W$ in week $k \in K$ (days that worker cannot work in week k due to holydays, fixed days-off or other unavailability reason)
S	set of all skills $s \in S$
L	ordered set of all competence levels among all skills, where $l = 1$ represents the highest level and $l = \|L\|$ represents the lowest level
$S_w \subset S$	set of skills of worker $w \in W$; for each $s \in S_w$, parameter $l_{ws} \in L$ indicates the competence level of worker $w \in W$ in skill $s \in S_w$
H_{wd}	set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$
α_{dts}	minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
β_{dtsl}	minimum number of workers required on day $d \in D$, timeslot $t \in T_d$, skill $s \in S$ and competence level $l \in L$
n_{wk}	number of working days that must be assigned to worker $w \in W$ in week $k \in K$ (given by the number of open days in week k minus the number of unavailable days of worker w in week k minus the number of preferred days-off of worker w in week k)

Variables:

x_{wdh}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$
x'_{wd}	binary variable that, when equal to 1, indicates that worker $w \in W$ works on a day $d \in D_{wk}$ (i.e., a preferable day-off of worker $w \in W$)
y_{wdts}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$
y'_{wdtsl}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ and competence level $l \in L$

z_{dts} integer variable with the number of workers below the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

z'_{dtsl} integer variable with the number of workers below the required value β_{dtsl} on day $d \in D$, timeslot $t \in T_d$, skill $s \in S$ and competence level $l \in L$

ObjectiveFunction1(α_{dts}): **trying to fulfil as much as possible the minimum number α_{dts} of workers in all days, timeslots and skills**

$$\text{Minimize} \sum_{d \in D_o} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \qquad (1)$$

The aim is to minimize the sum of the number of workers below the required value α_{dts} in all open days $d \in D_o$, timeslots $t \in T_d$ and skills $s \in S$.

ObjectiveFunction2(s, l, β_{dtsl}): **for a given skill s and competence level l, trying to fulfil as much as possible the minimum number β_{dtsl} of workers in all days and timeslots**

$$\text{Minimize} \sum_{d \in D_o} \sum_{t \in T_d} z'_{dtsl} \qquad (1')$$

The aim is to minimize the sum of the number of workers below the required value β_{dtsl} in all open days $d \in D_o$ and timeslots $t \in T_d$ for a given skill $s \in S$ and competence level $l \in L$.

ObjectiveFunction3: **trying to obtain the best possible average competence level**

$$\text{Minimize} \sum_{w \in W} \sum_{d \in D_o} \sum_{t \in T_d} \sum_{s \in S} \sum_{l \in L} (l \times y'_{wdtsl}) \qquad (1'')$$

The aim is to minimize the average competence level assigned to all workers $w \in W$ in all open days $d \in D_o$, all timeslots $t \in T_d$ and all skills $s \in S$ assuming linear competence level weights (i.e., competence level l has weight l).

ObjectiveFunction4: **trying to assign as much as possible the preferable days-off**

$$\text{Minimize} \sum_{w \in W} \sum_{d \in D_o} x'_{wd} \qquad (1''')$$

The aim is to minimize the number of preferable days-off of all workers that are assigned as working days.

Constraints:

$$\sum_{h \in H_{wd}} x_{wdh} \leq 1 \qquad , w \in W, d \in D_o \qquad (2)$$

With constraints (2), each worker $w \in W$ can be assigned on each open day $d \in D_o$ with at most one working daily assignment $h \in H_{wd}$.

$$\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} \delta_{wdht}\, x_{wdh} \qquad , w \in W, d \in D_o, t \in T_d \qquad (3)$$

With constraints (3), each worker $w \in W$ assigned on open day $d \in D_o$ with a working daily assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$ must be assigned in timeslot t with one of its skills S_w.

$$\sum_{d=d'}^{d'+5} \sum_{h \in H_{wd}} x_{wdh} \leq 5 \qquad , w \in W, d' \in \bar{D} \qquad (4)$$

With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive open days. (NOTE: In this version of the constraints, since there are close days, the constraints are required only for the sequences of 6 consecutive open days)

$$z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts} \qquad , d \in D_o, t \in T_d, s \in S \qquad (5)$$

With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each open day $d \in D_o$, each timeslot $t \in T_d$ and each skill $s \in S$.

$$y'_{wdtsc} = \begin{cases} y_{wdts} & , l = l_{ws} \\ 0 & , l \neq l_{ws} \end{cases} \qquad , w \in W, d \in D_o, t \in T_d, s \in S_w \qquad (6)$$

With constraints (6), each worker $w \in W$ that works in open day $d \in D_o$ in time slot $t \in T$ with skill $s \in S$ is assigned with its competence level l_{ws} and the variables associated to all other competence levels are set to zero.

$$z'_{dtsl} + \sum_{w \in W} y'_{wdtsl} \geq \beta_{dtsl} \qquad , d \in D_o, t \in T_d, s \in S, l \in L \qquad (7)$$

With constraints (7), the value of variable z'_{dtsl} is at least the number of workers below the required value β_{dtsl} for each open day $d \in D_o$, each timeslot $t \in T_d$, each skill $s \in S$ and each competence level $l \in L$. (NOTE: Constraints (7) are required only if ObjectiveFunction2 is used).

$$\sum_{h \in H_{wd}} x_{wdh} = 0 \qquad , w \in W, k \in K, d \in U_{wk} \qquad (8)$$

With constraints (8), each worker $w \in W$ cannot be assigned with any working daily assignment in the days that he is unavailable on each week.

$$\sum_{d \in D_k} \sum_{h \in H_{wd}} x_{wdh} = n_{wk} \qquad , w \in W, k \in K \qquad (9)$$

With constraints (9), each worker $w \in W$ must be assigned on each week $k \in K$ with n_{wk} working days.

$$x'_{wd} = \begin{cases} \sum_{h \in H_{wd}} x_{wdh} & , d \in D_{wk} \\ 0 & , d \notin D_{wk} \end{cases} \qquad , w \in W, k \in K, d \in D_w \qquad (10)$$

With constraints (10), variable x'_{wd} is set to 1 when worker $w \in W$ works on a day $d \in D_{wk}$ (i.e., a preferable day-off of worker $w \in W$), and is set to 0 in all other cases. (NOTE: Constraints (10) are required only if ObjectiveFunction4 is used).